Table 2: ResNets Configuration.

ResNet18		ResNet34		ResNet50		ResNet101		ResNet152	
conv3-64		conv3-64		conv1-64		conv1-64		conv1-64	
conv3-64	$\times 2$	conv3-64	$\times 3$	conv3-64	$\times 3$	conv3-64	$\times 3$	conv3-64	$\times 3$
				conv1-256		conv1-256		conv1-256	
conv3-128		conv3-128		conv1-128		conv1-128		conv1-128	
conv3-128	$\times 2$	conv3-128	$\times 4$	conv3-128	$\times 4$	conv3-128	$\times 4$	conv3-128	$\times 8$
				conv1-512		conv1-512		conv1-512	
conv3-256		conv3-256		conv1-256		conv1-256		conv1-256	
conv3-256	$\times 2$	conv3-256	$\times 6$	conv3-256	$\times 6$	conv3-256	$\times 23$	conv3-256	$\times 36$
				conv1-1024		conv1-1024		conv1-1024	
conv3-512		conv3-512		conv1-512		conv1-512		conv1-512	
conv3-512	$\times 2$	conv3-512	$\times 3$	conv3-512	$\times 3$	conv3-512	$\times 3$	conv3-512	$\times 3$
				conv1-2048		conv1-2048		conv1-2048	

Table 3: 10-layer CNN Configuration.

10-layer CNN
conv3-16 BN ReLu$\times 2$
conv3-32 (stride=2)BN ReLu
conv3-32 BN ReLu $\times 2$
conv3-64 (stride=2)BN ReLu
conv3-64 (valid padding)
conv1-64 BN ReLu
Global average pooling

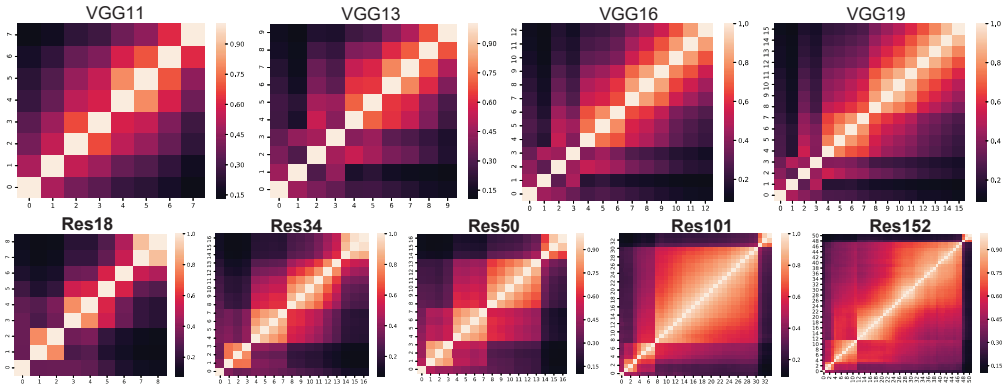


Figure 8: The similarity between the layers of VGGNets and ResNets on CIFAR100.

grids of ResNets show higher similarity between each other, while they are less similar on ImageNet. This phenomenon is more clear in VGGNets. For VGGNets, the similarity is very low even within a block on ImageNet.

We also conduct the same experiment using CKA on CIFAR10, CIFAR100, and ImageNet. The results are shown in Fig. 10, Fig. 11, and Fig. 12. CKA also demonstrates grids on the similarity matrix, however, the grids are less clear than GBS, especially on CIFAR100 and ImageNet. For VGGNets on CIFAR10 and CIFAR100, CKA does not show clear grids corresponding to the blocks in VGGNets, while GBS does.

D Extensibility of GBS

In our work, we adopt LSim to measure the similarity of graphs that correspond to hidden representations. There are other graph similarity measurements in addition to LSim, e.g., degree correlation.

The degree of a node is defined as the number of edges connected to it. If we arrange the degrees of $\mathcal{V} \in \mathcal{G}$ into a sequence $\mathbf{k}$, then $\mathbf{k}$ is called the degree sequence of $\mathcal{G}$. Assume the degree sequence of layer i is $\mathbf{k}_i$, which is a $1 \times N$ vector, then we can use correlations like Pearson to measure the

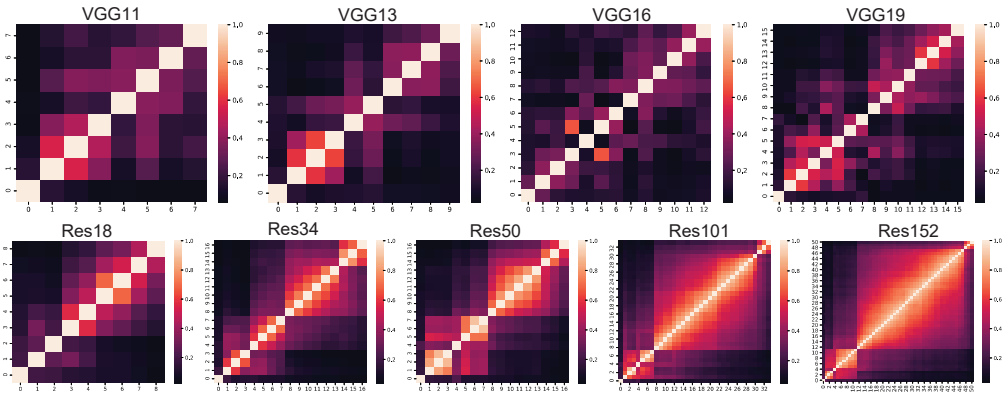


Figure 9: The similarity between the layers of VGGNets and ResNets on ImageNet.

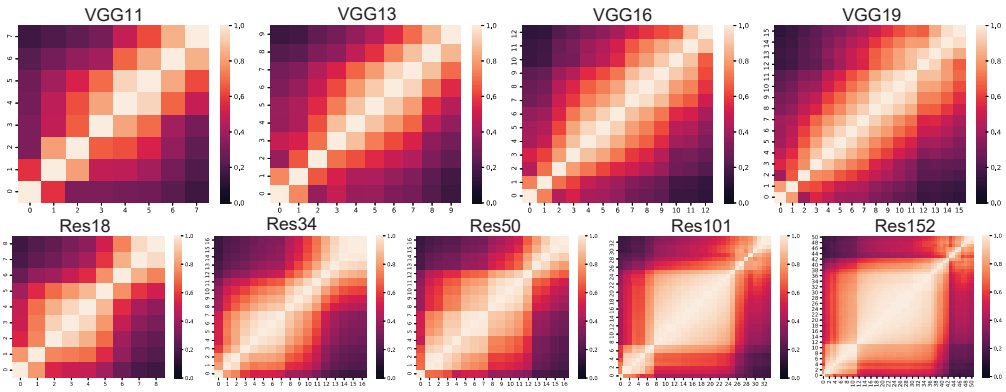


Figure 10: The CKA similarity between the layers of VGGNets and ResNets on CIFAR10.

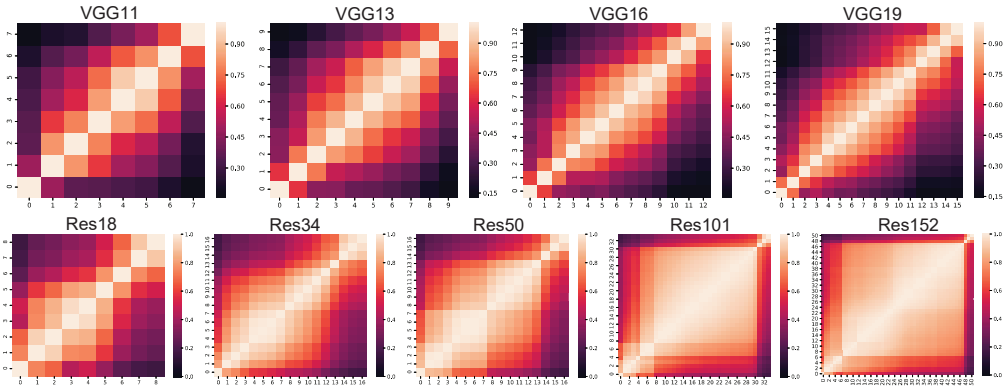


Figure 11: The CKA similarity between the layers of VGGNets and ResNets on CIFAR100.

similarity. The degree correlation of layer i and j is

$$\rho_{i,j} = \frac{\mathbb{E}[(\mathbf{k}_i - \mu_i)(\mathbf{k}_j - \mu_j)]}{\sigma_i \sigma_j}, \tag{11}$$

where μ_i is the mean value of $\mathbf{k}_i$ and σ_i is the variance of $\mathbf{k}_i$.

We demonstrate the ablation analysis result of GBS using degree sequence and Pearson similarity, as shown in Fig. 13. The experiment setting is the same with the cosine similarity. Since the Pearson correlation varies from -1 to $+1$, we take the absolute value of the result as the similarity score. Compared with cosine similarity, Pearson correlation shows greater volatility within the same degree range. The accuracy is also lower under the same graph size.

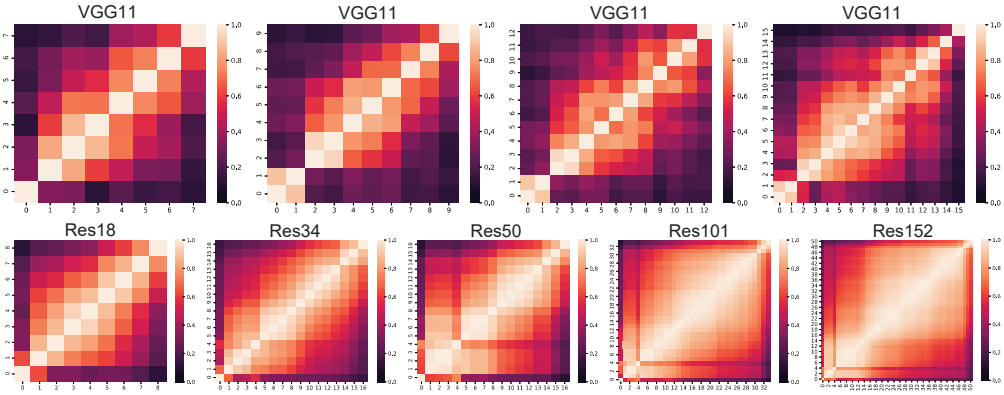
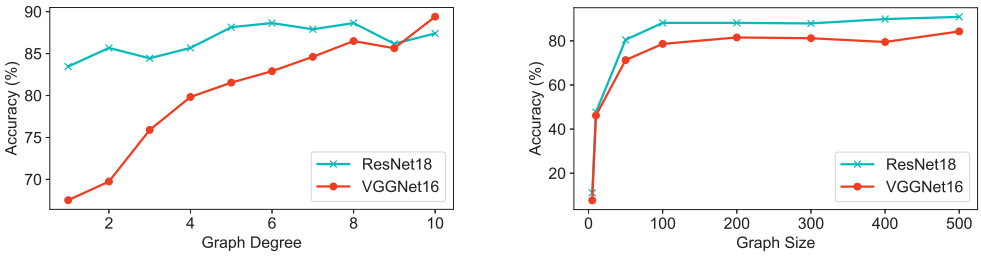


Figure 12: The CKA similarity between the layers of VGGNets and ResNets on ImageNet.



(a) (b)

Figure 13: Sanity check accuracy using Pearson correlation with VGGNet16 and ResNet18 on CIFAR10.